
Data Structures

BS (CS) _Fall_2024

Lab_06 Tasks



Learning Objectives:

1. Single Linked List

Lab Tasks

Submission Instructions

1. Create a separate **.h** and **.cpp** file for each class.
2. Now create a new folder with name *ROLLNO_SEC_LAB01* e.g. **i23XXXX_A_LAB01**(Also name your **cpp** file like this.)
3. Move all of your files to this newly created directory and compress it into a **.zip file**. Return the test file as well.
4. Now you have to submit this zipped file on Google Classroom.
5. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.
6. Plagiarism or AI generated code in the Lab Task will result in **zero** marks in the whole category.

Question 1

Company ABC have hired you as their lead developer; and the first task assigned to you by your manager is to store the employee data efficiently so that it will be easy for the company to add new employee at any time.

For the time being, you have selected to proceed with the single Linked List implementation to store the Company employee data.

Following are the details that your company wants you to store about an employee;

1. Name
2. NIC
3. Date of Joining
4. Gender
5. Marital Status
6. Salary
7. Designation
8. Employee Scale
9. Department

You are required to create a proper OOP hierarchy to store all of the data in a Single Linked list with following features;

1. Add Employee
2. Search Employee With NIC
3. List of Employee Who have a certain joining date
4. Employees in ascending sort order based on their salary
5. Employees from a certain department
6. Employees with a certain department and designation
7. Edit the record of an Employee with a certain NIC (Editable Data: Salary, Marital Status, Designation, Employee Scale, Department)

Question 2

Design a class for Singly linked List with data members **data** and **next** pointer. Your class must provide implementation for the following

- A default constructor for Node
- A parameterized constructor
- Getters for data and next
- Setters for data and next

Design a LinkedList class with a pointer of type Node. This will carry the address of head initially, later it will be required for traversal.

Your class must provide implementation for the following

- A default constructor
- Insert function to insert data item. For every new data element to be inserted, you will have to reach up to the end point of LL and then perform insertion
- Insert to head function to insert data element at the start of your LL
- Design isEmpty to check if there is no data element in the LL
- Why don't you need to design isFull?
- A Search function to search for any element provided by user
- Update function to update/replace one data element with another
- A function Insert at index that will count data elements in the LL and then place a new data element in the specified location. To know exactly after which node we have to make insertion, we need count the elements and that will help us to identify index (index doesn't exist actually, but we are using this term to get to know the location for insertion)
- A function Delete to delete the specified data element, for this you will have to save the address of the next node that appears after the one to be deleted. Saving the address will help you after deleting the element to concatenate the list with the elements appearing after the element that is supposed to be deleted.
- A function to print the data of LL